



Designing Authentic Industry-Engaged Assessment for Professional Competence in Business Intelligence

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INDUSTRIAL ENGINEERING

Academia

Aims to prepare students for professional practice, but offers limited guidance on how to operationalize this in course design and assessment

Industry

Expects credible, reproducible results supported by clear reasoning and usable for decision-making

Underlying mismatch

What counts as “good work” differs between academic and professional contexts

What prompted the redesign

Observed in an industry-facing course

-  Industry partners valued collaboration – but would not use the results
-  Students reported a very heavy and uneven workload

Core question

How do we design courses so that student work is **authentic in practice**, while still supporting **meaningful learning**?

Overview

- ▶ 6 ECTS course
 -  advanced undergraduate
 -  early graduate
- ▶ 10–15 students, two teams

What is BI?

Data-driven decision support: combining data analysis, systems, and communication to inform organizational decisions.

Goal

Strengthen industrial relevance and professional competence

Sustained industrial context

A single dataset and domain used across the entire semester

Industry partner role

- ▶ problem framing
- ▶ milestone discussions
- ▶ final evaluation

Dual context

Academic and industry perspectives coexist



Design focus

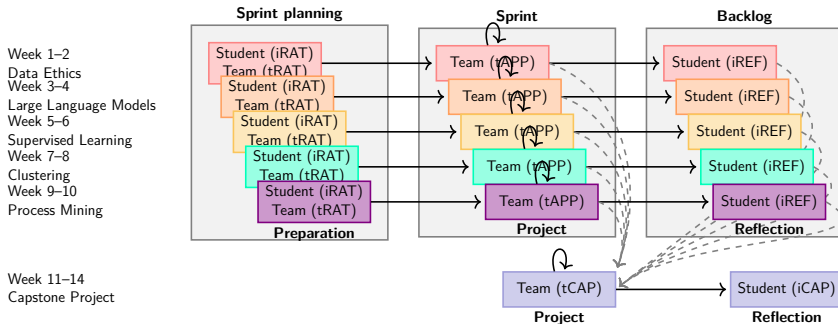
- ▶ limited set of competences (LOUIS)
- ▶ sustained industry context
- ▶ iterative development toward capstone

LOUIS framework

Learning Outcomes in University for Impact on Society
Focus on a small set of competences (max 3)

Civic engagement	Creative thinking	Critical thinking	Ethical reasoning
Global learning	Information literacy	Inquiry and analysis	Integrative learning
Intercultural knowledge & competence	Foundations for life-long learning	Oral communication	Problem solving
Quantitative literacy	Reading	Teamwork	Written communication

Pedagogical structure



 Team-Based Learning (TBL):

-  iRAT (individual readiness)
-  tRAT (team readiness)
-  tAPP (team application)
-  iREF (individual reflection)

 Flipped classroom

 Iterative modules leading into capstone

Core mechanism: assessment as development

☰ One rubric, many iterations

Same assessment rubric reused across all modules and the capstone

Standards fixed; performance expectations scaled during learning

☑ Assessment criteria

- ▶ analytical soundness
- ▶ transparency
- ▶ reproducibility
- ▶ stakeholder relevance

▶▶ Developmental shift

Feedback is **carried forward**
— *not* reset between tasks

Pull Requests as assessment architecture

Changes are proposed, reviewed, discussed, and integrated – or explicitly deferred

Structured accumulation

- ▶ same rubric reused
- ▶ prior feedback revisited
- ▶ unresolved issues explicitly carried forward

Enabled by tooling

-  version control
-  persistent work traces

Instructor

- ▶ method
- ▶ implementation

Industry partner

- ▶ domain relevance
- ▶ decision context

Student

- ▶ knowledge transfer through PRs
- ▶ defending decisions
- ▶ critiquing peer work

Progression over time

1. Align within teams
2. Cross-team review
3. Industry validation

Audience shift

From internal review
to external scrutiny

Industry threshold

Credibility is fragile: early issues can undermine trust

What is evaluated

- ▶ reproducibility
- ▶ transparency
- ▶ clear assumptions

Shift in communication

Build trust first
– then invite inspection

Team-level assessment

- ▶ analytical quality
- ▶ coherence
- ▶ relevance

Individual-level assessment

- ▶ visible artifacts
- ▶ documented contributions
- ▶ structured reflection

Design goal

Accountability without fragmenting teamwork

Designed into the course architecture—not added after the fact



Students & graduates

- ▶ high engagement without attendance requirements
- ▶ transfer of reproducible workflows
- ▶ stronger documentation practices

Academic and industry perspectives



more ambitious, student-driven projects



improved coherence and credibility

⚡ What becomes visible

The design makes certain challenges explicit

- ▶ giving and receiving critical feedback
- ▶ making work visible and reviewable
- ▶ managing workload in a structured workflow

Transferable principles

For CDIO-aligned, industry-engaged courses

- ▶ sustained industry context (no repeated onboarding)
- ▶ assessment drives iteration (not just evaluation)
- ▶ contributions are visible and reviewable
- ▶ feedback accumulates across the course
- ▶ students adapt to different audiences
- ▶ tooling supports – but does not define – the design

Current limitations

- ▶ depends on student readiness
- ▶ scalability considerations
- ▶ domain-specific transferability

Design refinements

- ▶ earlier preparation for professional tooling
- ▶ improved stakeholder interaction
- ▶ structured team-building to accelerate trust in large groups
- ▶ scaling across courses and partners

Questions?



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